

Sumit Kumar Kumawat

📍 Sikar, Rajasthan

☎ +91 8003207615 | ✉ 2023uec1484@mnit.ac.in | 🔗 LinkedIn | 🐙 GitHub | ⚡ LeetCode | 🛠️ GFG

Education

- **Malaviya National Institute Of Technology, Jaipur** **2023 – 2027**
B.Tech - Electronics and Communication engineering
CGPA: 7.89 Jaipur, Rajasthan

Projects

CareerSync – AI Career Assistant | *React.js, Node.js, Express.js, MongoDB, RAG, LangChain, Qdrant*

🐙 GitHub | 🚀 Live Demo

- Developed **CareerSync AI**, a MERN-based AI career assistant platform with JWT authentication and role-based access control, supporting secure user and recruiter workflows.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline using LangChain, Gemini, and Qdrant, reducing LLM context size by **60–70%** through semantic resume retrieval and vector-based search.
- Implemented an AI-powered job recommendation engine that predicts candidate roles from resume data and matches users with relevant job opportunities.
- Developed scalable RESTful APIs using Node.js, Express.js, FastAPI, and MongoDB, and implemented conversational memory to maintain context across multi-turn interactions.
- Conducted Grafana k6-based load testing with up to **500 concurrent virtual users**, validating application scalability, stability, and reliable request handling under high-concurrency workloads.

Speech Enhancement Using Deep Learning (U-Net) | *Python, PyTorch, U-Net, STFT*

🐙 GitHub

- Designed and trained a U-Net encoder-decoder network with skip connections to suppress background noise and enhance single-channel speech signals.
- Developed an end-to-end speech enhancement pipeline using STFT-based spectrogram generation, magnitude spectrogram prediction, and waveform reconstruction via ISTFT.
- Implemented the model in PyTorch using L1 reconstruction loss, Adam optimization, and spectrogram-based learning for robust speech recovery.
- Achieved improvements of **+4.85 dB SNR**, **+0.747 PESQ**, and **+0.052 STOI**, significantly improving speech quality and intelligibility.

Air Traffic Control Management System | *C++, OOPS, OS, Multithreading, DSA, STL*

🐙 GitHub

- Built an Air Traffic Control System simulating airport operations with runway and gate allocation.
- Designed priority-based flight scheduling with emergency handling using STL and custom comparators.
- Used multithreading, mutexes, condition variables, and SQLite for synchronization and flight logging.
- Conducted stress testing with **1000+** Flights, evaluating scheduling latency and concurrent system performance.

Technical Skills

- **Languages:** C/C++, JavaScript, Python, SQL
- **Web Technologies:** HTML, CSS, React.js, Node.js, Express.js, FastAPI, REST APIs
- **Databases:** MongoDB, MySQL, SQLite, Qdrant (Vector DB)
- **Core CS:** Data Structures & Algorithms, OOPS, DBMS, Operating Systems
- **LLM & AI:** LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering
- **Libraries:** PyTorch, Scikit-learn, NumPy, Pandas
- **Developer Tools:** Git, GitHub, Postman, VS Code, Jupyter Notebook, Grafana k6, Vercel, Render

Coding Platforms

- Solved 500+ DSA problems across LeetCode and GeeksforGeeks.
- Completed Coding Ninjas Ninja Slayground 2.0 (21-Day Coding Challenge).

Leadership & Activities

- Executive Member, Music Club – organized musical events and team activities.